

 Text Functions & Extraction XLOOKUP & INDEX/MATCH

INSTANT

 Data Analysis Diploma • Session 05

ADVANCED FUNCTIONS

Text Functions · Lookup Functions · Dynamic Arrays · Power Query
Fundamentals

6 Goals**Core Outcomes**

Text • Lookups • Power Query

44 Slides**Curriculum Depth**

Formulas & Power Query

Visual Labs**Hands-on Practice**

Lookup Simulator & PQ

Start Session 05 → Dynamic Array Formulas Power Query ETL Pipelines

— SESSION COMPETENCIES

What You Will Master Today

By the end of this session, you will confidently apply these advanced Excel and ETL capabilities:



01

Master Text Extraction

Extract, clean, and reformat strings using LEFT, RIGHT, MID, LEN, FIND, and SUBSTITUTE.



02

Advanced Lookups

Connect datasets seamlessly with XLOOKUP, VLOOKUP, HLOOKUP, and INDEX/MATCH.



03

Dynamic Array Formulas

Build self-updating reports using FILTER, UNIQUE, SORT, and SORTBY functions.



04

Power Query Fundamentals

Automate repetitive data cleaning tasks without writing a single line of code.



05

ETL Data Pipeline

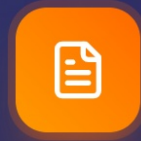
Import, clean, transform, and load messy data into analytical models with 1-click refresh.



06

Real Business Scenarios

Construct automated sales reporting dashboards, HR directories, and inventory logs.



PART 1

Excel Text Functions

Extracting, Parsing, Standardizing, and Combining Text Data like a Professional Analyst

— STRING PARSING

LEFT & RIGHT Functions

Extract characters from the beginning or end of a text string.



KEY CONCEPTS & SYNTAX

- LEFT extracts a specified number of characters from the start of text.
- RIGHT extracts characters from the end of text.
- Essential for splitting product codes, IDs, zip codes, or names.
- Syntax: =LEFT(text, num_chars) | =RIGHT(text, num_chars)
- Common Mistake: Specifying more characters than exist in the source cell.

EXCEL FORMULA SYNTAX

```
=LEFT(A2, 3)
```

```
=RIGHT(A2, 4)
```

REAL BUSINESS EXAMPLE

Extract product category code "ELEC" and product serial "2045" from SKU "ELEC-2045".

— MIDDLE EXTRACTION & LENGTH

MID & LEN Functions

Extract substring from the middle of text and measure string length.



KEY CONCEPTS & SYNTAX

- MID extracts characters from inside a text string starting at a specific position.
- LEN returns the total character count of a text string.
- MID requires: text, start_num, and num_chars.
- LEN is ideal for data validation (e.g. verifying 10-digit phone numbers or 14-digit IDs).
- Common Mistake: Incorrect start position skips or grabs wrong characters.

EXCEL FORMULA SYNTAX

```
=MID(A2, 6, 4)
```

```
=LEN(A2)
```

REAL BUSINESS EXAMPLE

Extract the year "2045" from SKU "ELEC-2045-A" and validate ID length.

— TEXT CONCATENATION

CONCAT & TEXTJOIN Functions

Combine multiple text strings into a single formatted value.



KEY CONCEPTS & SYNTAX

- CONCAT joins text from multiple cells or ranges without delimiters.
- TEXTJOIN joins text with a chosen delimiter (comma, space, dash).
- TEXTJOIN automatically ignores empty cells when ignore_empty is set to TRUE.
- Great for assembling full names, complete mailing addresses, or combined tags.
- Common Mistake: Forgetting delimiter argument or quotes in TEXTJOIN.

EXCEL FORMULA SYNTAX

```
=CONCAT(B2, C2)
```

```
=TEXTJOIN(", ", TRUE, D2:D5)
```

||| REAL BUSINESS EXAMPLE

Combine First Name and Last Name columns into one Full Name, or concatenate address lines.

— TEXT FORMATTING

UPPER, LOWER & PROPER

Standardize letter casing across inconsistent customer and product records.



KEY CONCEPTS & SYNTAX

- UPPER converts all characters in a text string to UPPERCASE.
- LOWER converts all characters to lowercase.
- PROPER capitalizes the first letter of each word and lowers all other letters.
- Standardizes inconsistent manual data entry from web forms or CRM exports.
- Common Mistake: Applying to numeric values or expecting automatic live cell updates.

EXCEL FORMULA SYNTAX

`=UPPER(E2)`

`=LOWER(E2)`

`=PROPER(E2)`

REAL BUSINESS EXAMPLE

Standardize customer names entered inconsistently ("jOHn sMItH" -> "John Smith").

— WHITESPACE REMOVAL

TRIM Function

Remove excess spaces from text while keeping single spaces between words.



KEY CONCEPTS & SYNTAX

- Removes all leading spaces, trailing spaces, and multiple spaces between words.
- Keeps a single space character between words for proper formatting.
- Essential for cleaning imported CSV, web data, or copied database extracts.
- Fixes mysterious lookup failures caused by invisible trailing spaces.
- Common Mistake: Assuming TRIM removes non-breaking spaces (CHAR160) without SUBSTITUTE.

EXCEL FORMULA SYNTAX

```
=TRIM(F2)
```

REAL BUSINESS EXAMPLE

Clean up messy email addresses copied from webforms ("john.doe@email.com " -> "john.doe@email.com").

— POSITION SEARCH

FIND Function

Locate the exact character position of a substring within text.



KEY CONCEPTS & SYNTAX

- Locates the starting position of a target substring within a text string.
- Case-sensitive search function (unlike SEARCH which is case-insensitive).
- Returns a numeric position integer (e.g. 5 means the character is at position 5).
- Frequently combined with MID or LEFT for dynamic, variable-length text extraction.
- Common Mistake: Text not found returns a #VALUE! error (wrap with IFERROR or ISNUMBER).

EXCEL FORMULA SYNTAX

```
=FIND("-", G2)
```

REAL BUSINESS EXAMPLE

Locate the dash position within an order reference number ("ORD-99214") to dynamically split IDs.

— STRING REPLACEMENT

SUBSTITUTE Function

Replace specific text within a string programmatically.



KEY CONCEPTS & SYNTAX

- Replaces existing text with new text in a text string.
- Can target all occurrences or a specific occurrence instance if needed.
- Ideal for correcting formatting errors, changing date separators, or stripping unwanted characters.
- Different from Find & Replace tool because it updates dynamically via formula.
- Common Mistake: Case sensitivity causes missed replacements if case doesn't match.

EXCEL FORMULA SYNTAX

```
=SUBSTITUTE(H2, "/", "-")
```

```
=SUBSTITUTE(H2, " ", "", 1)
```

REAL BUSINESS EXAMPLE

Standardize date separators across imported reports
("12/05/2024" -> "12-05-2024").

— NESTED FORMULAS

Combining MID + FIND + LEN

Extracting variable-length sub-strings dynamically when positions change.

1. FIND DELIMITER

FIND("-", A2) locates the hyphen position in string "PROD-88219-US".

`FIND("-", A2) -> 5`

2. MID EXTRACTION

MID starts at position 6 (5 + 1) and extracts 5 digits.

`MID(A2, 6, 5) -> "88219"`

3. WRAP WITH TRIM

TRIM ensures no hidden spaces remain after extraction.

`TRIM(MID(A2, FIND("-", A2)+1, 5))`

PRACTICAL APPLICATION: Extract product serial numbers of varying lengths from non-standardized invoice codes.

— LAB 01

Interactive Text Cleaning Studio

Test text functions live on messy raw datasets and observe output transformations.

1. SELECT INPUT SAMPLE

Messy SKU

" ELEC-2045-US "

Unformatted Name

"jOHn.sMItH@company.COM "

Invoice Code

"ORD/2024/9912"

2. SELECT FUNCTION

TRIM

UPPER

LOWER

PROPER

LEFT (4)

RIGHT (4)

SUBSTITUTE (/) to (-)

3. LIVE EXCEL TRANSFORMATION RESULT

RAW CELL VALUE (A2)

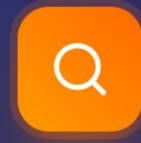
" ELEC-2045-US "

EVALUATED FORMULA

=TRIM(A2)

TRANSFORMED RESULT OUTPUT:

"ELEC-2045-US"



PART 2

Lookup & Reference Functions

Searching, Connecting, and Retrieving Data Across Tables with VLOOKUP, HLOOKUP, XLOOKUP & INDEX/MATCH

— CLASSIC LOOKUP

VLOOKUP Function

Search for a value in the first column of a table and return a value in the same row from another column.



KEY CONCEPTS & SYNTAX

- Searches for a value in the FIRST column of a table range.
- Returns a matching value from a specified column index number.
- Requires setting range_lookup to FALSE (or 0) for exact matches.
- Widely used across business models, but restricted to left-to-right lookups only.
- Common Mistake: Forgetting FALSE/0 defaults to TRUE (approximate match), returning wrong data!

EXCEL FORMULA SYNTAX

```
=VLOOKUP(I2, ProductTable, 3, FALSE)
```

REAL BUSINESS EXAMPLE

Look up product unit price using Product ID from a master pricing table.

— ROW-BASED LOOKUP

HLOOKUP Function

Search for a value in the top row of a table and return a value in the same column from a specified row.



KEY CONCEPTS & SYNTAX

- Searches for a value in the FIRST row of a horizontal data table.
- Returns a matching value from a specified row index number.
- Used when data headers are arranged vertically along rows.
- Follows exact same match logic as VLOOKUP (set FALSE for exact match).
- Common Mistake: Attempting to use HLOOKUP on standard vertical datasets.

EXCEL FORMULA SYNTAX

```
=HLOOKUP(J2, SalesTable, 4, FALSE)
```

REAL BUSINESS EXAMPLE

Find quarterly revenue from a horizontally structured finance summary sheet.

— RECOMMENDED STANDARD

XLOOKUP Function

The modern, flexible replacement for VLOOKUP and HLOOKUP in Excel 365.



KEY CONCEPTS & SYNTAX

- Modern replacement that supersedes VLOOKUP, HLOOKUP, and INDEX/MATCH.
- Searches left, right, up, or down — in any direction without column index numbers.
- Defaults to exact match automatically (no need to type FALSE).
- Includes built-in "if_not_found" error handling parameter to prevent #N/A errors.
- Common Mistake: Trying to use XLOOKUP in older legacy versions of Excel (pre-2021/365).

EXCEL FORMULA SYNTAX

```
=XLOOKUP(K2, IDRange, NameRange, "Not Found")
```

REAL BUSINESS EXAMPLE

Retrieve employee names using employee IDs, even when ID column is to the right of Name column.

— POWERFUL PAIR

INDEX & MATCH Combination

The classic high-performance alternative for complex lookups in any version of Excel.



KEY CONCEPTS & SYNTAX

- INDEX returns a value at a given row and column position in a range.
- MATCH finds the row position number of a search value in a column.
- Combined, MATCH finds the row number and INDEX pulls the corresponding data.
- Can look up in any direction (left or right) and does not break when columns are inserted.
- Common Mistake: Mismatched range sizes between INDEX return range and MATCH lookup range.

EXCEL FORMULA SYNTAX

```
=INDEX(NameRange, MATCH(L2, IDRange, 0))
```

REAL BUSINESS EXAMPLE

Retrieve a project manager's name based on Project ID when columns are reorganized.

METHOD EVALUATION

Lookup Function Comparison

Choose the right lookup methodology based on data structure and Excel environment.

Feature	VLOOKUP	HLOOKUP	XLOOKUP ★	INDEX & MATCH
Lookup Direction	Left-to-Right only	Top-to-Bottom only	Any direction (Left/Right/Up/Down)	Any direction
Match Default	Approximate (Requires FALSE)	Approximate (Requires FALSE)	Exact Match by Default	Requires 0 for Exact Match
Column Insertion Safety	Breaks (Static column index)	Breaks (Static row index)	Safe (Dynamic array reference)	Safe (Dynamic array reference)
Built-in Error Handling	No (Requires IFERROR)	No (Requires IFERROR)	Yes (if_not_found argument)	No (Requires IFERROR)
Excel Version	All Excel Versions	All Excel Versions	Excel 365 / 2021+	All Excel Versions

BUSINESS ADVICE: A financial services firm migrates legacy VLOOKUP reports to XLOOKUP for speed and reliability.

CLEAN REPORTS

Shielding Lookups with IFERROR

Prevent ugly #N/A, #VALUE!, and #REF! errors from spoiling executive dashboards.



KEY CONCEPTS & SYNTAX

- IFERROR catches any formula error and replaces it with a clean fallback value or text.
- IFNA specifically catches #N/A missing match errors while letting other calculation errors surface.
- Prevents downstream calculations (SUM, AVERAGE) from breaking due to a single missing match.
- Essential when looking up newly added products or customer IDs that may not exist yet.
- Common Mistake: Wrapping formulas with IFERROR too early without fixing broken formula syntax.

EXCEL FORMULA SYNTAX

```
=IFERROR(VLOOKUP(A2, ProductTable, 3, FALSE), "Not Found")
```

```
=IFNA(XLOOKUP(A2, IDRange, PriceRange), 0)
```

REAL BUSINESS EXAMPLE

Replace "#N/A" errors with "Item Discontinued" or 0 in sales invoice lookup tables.

— TWO-WAY MATCHING

Two-Way Matrix Lookup (INDEX + 2 MATCHes)

Looking up values across both rows and columns simultaneously in a matrix.

1. ROW MATCH

MATCH(Region, RegionList, 0) finds the row number for "North".

```
MATCH("North", A2:A10, 0) -> Row 3
```

2. COLUMN MATCH

MATCH(Month, MonthList, 0) finds the column number for "March".

```
MATCH("March", B1:M1, 0) -> Col 3
```

3. INDEX MATRIX FETCH

INDEX returns value at intersection of Row 3 and Col 3.

```
INDEX(B2:M10, 3, 3)
```

PRACTICAL APPLICATION: Retrieve monthly budget figures from a 2D matrix matching both Region and Month.

— SIMULATOR

Interactive Lookup Algorithm Simulator

Compare execution of VLOOKUP vs XLOOKUP vs INDEX/MATCH on a live product database.

1. SELECT TARGET ID

P-101

P-102

P-103

P-104

2. SELECT LOOKUP METHOD

XLOOKUP

VLOOKUP

INDEX/MATCH

LIVE PRODUCT TABLE DATA

MATCHING ROW #2

Row #	Product ID (Col A)	Product Name (Col B)	Price (Col C)
1	P-101	MacBook Pro 16"	\$2,499
2	P-102	Dell XPS 15	\$1,899
3	P-103	iPad Pro 12.9"	\$1,099
4	P-104	Logitech MX Master	\$99

EXECUTED EXCEL FORMULA:

=XLOOKUP("P-102", A2:A5, C2:C5) -> \$1,899



PART 3

Dynamic Array Functions

Next-Generation Filtering, Sorting & Distinct Extraction with Modern Excel
Dynamic Engines

— AUTO-SPILL

FILTER Function

Extract rows that meet specified logical criteria dynamically.



KEY CONCEPTS & SYNTAX

- Returns an array of rows that satisfy one or more logical criteria.
- Dynamically spills results into adjacent empty cells automatically.
- No need for manual filter menus, copy-pasting, or complex array formulas.
- Updates live whenever source data or filter criteria change.
- Common Mistake: Criteria range length not matching data range height causes #VALUE! error.

EXCEL FORMULA SYNTAX

```
=FILTER(OrdersTable, Region="East")
```

```
=FILTER(OrdersTable, (Region="East")*(Sales>5000),  
"No Records")
```

||| REAL BUSINESS EXAMPLE

Instantly display all high-value orders placed from the "East" region.

— DUPLICATE STRIPPING

UNIQUE Function

Extract distinct values from a column or range automatically.



KEY CONCEPTS & SYNTAX

- Extracts unique, distinct values from a list or multi-column range.
- Removes duplicate entries automatically without altering source data.
- Ideal for building clean data validation drop-down lists dynamically.
- Combines seamlessly with SORT for organized, alphabetical outputs.
- Common Mistake: Expecting UNIQUE to modify the original source dataset.

EXCEL FORMULA SYNTAX

```
=UNIQUE(CustomerList)
```

```
=SORT(UNIQUE(CategoryColumn))
```

||| REAL BUSINESS EXAMPLE

Generate a clean list of unique customer names from a raw orders log of 10,000+ rows.

— AUTO-RANKING

SORT Function

Arrange data in ascending or descending order dynamically.



KEY CONCEPTS & SYNTAX

- Sorts the contents of a range or array based on a specified column index.
- Supports ascending (1) and descending (-1) sort directions.
- Dynamic — automatically re-sorts output when underlying source numbers update.
- Does not alter or touch original data rows.
- Common Mistake: Sorting only one column of a multi-column table without selecting full range.

EXCEL FORMULA SYNTAX

```
=SORT(SalesTable, 3, -1)
```

REAL BUSINESS EXAMPLE

Automatically rank sales representatives by revenue from highest to lowest.

— FLEXIBLE SORTING

SORTBY Function

Sort a range based on values in another column or external array.



KEY CONCEPTS & SYNTAX

- Sorts a range based on another corresponding range or array values.
- Allows sorting by columns that are not included in the final output array.
- Supports multiple sort levels (e.g. Sort by Department ascending, then Salary descending).
- Provides greater flexibility than standard SORT for custom matrix reports.
- Common Mistake: Mismatched array lengths between target range and sort-by range.

EXCEL FORMULA SYNTAX

```
=SORTBY(ProjectList, PriorityLevel, -1)
```

```
=SORTBY(EmpNames, DeptList, 1, SalaryList, -1)
```

REAL BUSINESS EXAMPLE

Sort project tasks by priority level score, rather than alphabetically by task name.

— POWER TRIAD

Combining **FILTER + UNIQUE + SORT**

Constructing powerful single-cell dynamic reports that update in real time.

1. FILTER DATA

`FILTER(CustomerNames, Status="Active")` returns active customer names.

```
FILTER(A2:A100, B2:B100="Active")
```

2. UNIQUE EXTRACTION

`UNIQUE` strips away duplicate customer orders to get distinct names.

```
UNIQUE(FILTER(...))
```

3. SORT ALPHABETICALLY

`SORT` arranges the unique active customer list alphabetically.

```
=SORT(UNIQUE(FILTER(A2:A100,
B2:B100="Active")))
```

PRACTICAL APPLICATION: Generate an automated, self-updating dropdown list of active unique client names.

— CASE STUDY 01

Dynamic Sales Reporting Dashboard

Combine XLOOKUP + FILTER + TEXTJOIN for live executive dashboard automation.

IMPLEMENTATION STEPS

- ✓ Combine XLOOKUP and FILTER for an auto-updating live sales dashboard.
- ✓ Pull product unit prices automatically into raw orders upon transaction.
- ✓ Filter live results dynamically by Region and Sales Representative.
- ✓ Use TEXTJOIN to build combined summary order logs for management.
- ✓ Deliver a live, self-updating report that eliminates monthly re-work.

BUSINESS IMPACT & OUTCOME

"Sales dashboard auto-updates instantly as new order rows are appended."

⚡ Zero manual copy-pasting required on recurring reports.

— CASE STUDY 02

HR Searchable Employee Directory

Clean names, look up departments, and extract distinct teams for HR management.

IMPLEMENTATION STEPS

- ✓ Use CONCAT / TEXTJOIN to build standardized full employee names.
- ✓ Use PROPER to correct inconsistent name capitalization from job applications.
- ✓ Use INDEX / MATCH to pull department and manager names from org charts.
- ✓ Use UNIQUE to list all distinct departments for filter controls.
- ✓ Build a clean, searchable HR directory for 300+ staff members.

BUSINESS IMPACT & OUTCOME

"HR team saves 5+ hours per week managing employee onboarding logs."

⚡ Zero manual copy-pasting required on recurring reports.

— CASE STUDY 03

Automated Warehouse Inventory Control

Filter low-stock items and rank by reorder urgency automatically.

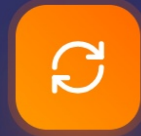
IMPLEMENTATION STEPS

- ✓ Use FILTER to isolate low-stock items where Stock < Minimum Threshold.
- ✓ Use SORT to rank items by reorder urgency score (highest priority first).
- ✓ Use SUBSTITUTE to clean messy SKU formatting across legacy warehouses.
- ✓ Use UNIQUE to list distinct product categories needing immediate restocking.
- ✓ Support faster, error-free warehouse decision-making and automated PO generation.

BUSINESS IMPACT & OUTCOME

"Warehouse logistics team gets a live low-stock priority list updated every morning."

⚡ Zero manual copy-pasting required on recurring reports.



PART 4

Power Query Introduction

Connecting, Cleaning, and Reshaping Data Without Writing Code

— FOUNDATION

What is Power Query?

A built-in Excel tool that connects, cleans, and reshapes data before it ever hits a worksheet.



Import Data

Connect to files (CSV, Excel), folders, databases & web APIs.



Clean Data

Remove errors, blank rows, extra spaces & inconsistent values.



Transform Data

Reshape rows & columns into the exact layout needed for analysis.



Load Data

Deliver clean, ready-to-use tables directly into Excel or Data Model.



DID YOU KNOW?

Power Query automatically records every transformation step you perform — so you can review, edit, or reuse the exact same cleaning process next time with a single click of Refresh!

— WHY POWER QUERY

Why Use Power Query?

The same clean-up work, two very different ways.

MANUAL CLEANING IN EXCEL

TIME CONSUMING

- ✗ Remove duplicates manually cell by cell ✗
- ✗ Fix date formats manually every month ✗
- ✗ Split text columns using Text-to-Columns wizard ✗
- ✗ Delete blank rows line by line ✗
- ✗ Change data types manually cell range ✗
- ✗ Re-do all work from scratch when new file arrives ✗

POWER QUERY AUTOMATION

1-CLICK REFRESH

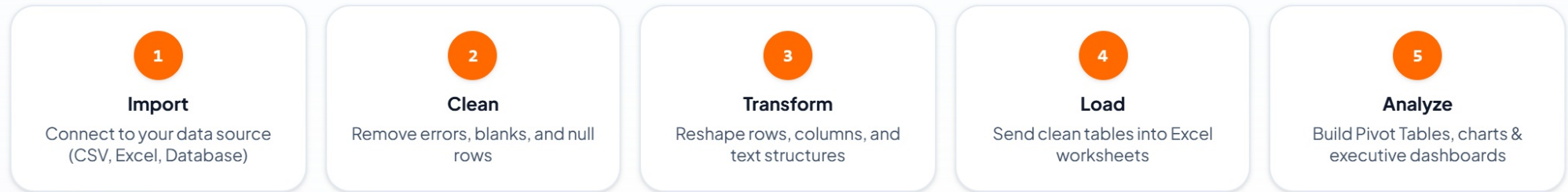
- ✓ Remove Duplicates in 1 Click ✓
- ✓ Fix Dates automatically with Data Types ✓
- ✓ Split Names dynamically by delimiter ✓
- ✓ Remove Blank Rows instantly across dataset ✓
- ✓ Enforce strict Data Types across columns ✓
- ✓ Click Refresh to re-run pipeline on new data! ✓

"Automate repetitive data prep work and focus on strategic analysis."

— 5 STAGES

Power Query Workflow

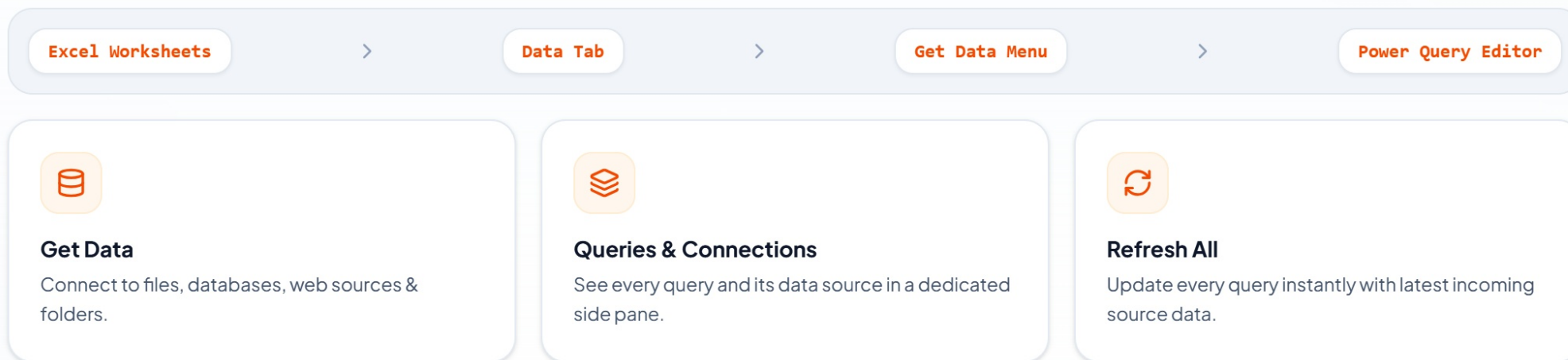
Five distinct stages from raw files to finished interactive dashboard analysis.



— NAVIGATING EXCEL

Where Can We Find Power Query?

Access Power Query directly inside standard Microsoft Excel without installing add-ins.



— INTERFACE TOUR

Power Query Editor Overview

One dedicated editor window, six essential functional areas to know.

Ribbon

Commands & transformation tools organized by tab.

Queries Pane

Lists every active query connection in workbook.

Data Preview Grid

Live interactive preview of current cleaned table.

Applied Steps

Sequential recorded history of every cleaning action.

Formula Bar

Displays the underlying Power Query M code for each step.

Column Headers

Data type indicators and quick column filter drop-downs.

— EVERYDAY TASKS

Common Cleaning Tasks in Power Query

Everyday cleaning actions, all just a few clicks away.



Remove Duplicates

Keep only unique data rows



Split Columns

Break one field into several columns



Trim Spaces

Clear extra leading & trailing spaces



Replace Values

Swap errors or bad text for right value



Change Data Types

Set text, number, date & boolean types



Filter Rows

Keep only specific data rows needed



Sort Data

Order rows ascending or descending



Rename Columns

Give headers clear, consistent names



Refresh Data

Re-run every step on new raw data

CORE DIFFERENCE

Excel Formulas vs Power Query

Two different jobs — both essential for data analysts.

Aspect	EXCEL FORMULAS	POWER QUERY
Scope	Cell-based calculations	Table-based transformations
Execution	Manual or row-by-row formulas	Automated ETL recorded steps
Data Refresh	Recalculates on cell edit	Refreshable pipeline on raw source
Primary Role	Calculations & Business Logic	Data Preparation & Cleaning

"Excel formulas calculate . Power Query prepares data. "

— PRACTICAL DEMO

Live Demo Step-by-Step Preview

What we will build together in the interactive lab, step by step.

1

Import File

Load messy raw CSV
customer export

2

Remove Blank Rows

Strip empty rows across
dataset

3

Fix Data Types

Convert text dates and
prices to numeric

4

Split Full Name

Separate names into First &
Last Name

5

Remove Duplicates

Deduplicate records by
Customer ID

6

Close & Load

Deliver clean table to Excel
worksheet

— SIMULATOR LAB

Interactive Power Query Transformation Simulator

Watch a messy raw customer table transform step-by-step through Power Query Applied Steps.

APPLIED STEPS PANE

1. Source (Raw Data)

2. Filtered Rows

3. Trimmed Text

4. Removed Duplicates

ACTION PERFORMED:

Import Raw CSV File

LIVE DATA PREVIEW GRID

3 CLEAN ROWS

Customer ID	Full Name	Join Date	Status
101	john smith	2024/01/15	null
102	MARY JOHNSON	2024-01-16	null
101	john smith	2024/01/15	null

Previous Step

Next Step

— DATA TYPES

Data Types & Error Handling in Power Query

Enforce strict column data types to guarantee calculation accuracy.

Data Types & Error Handling in Power Query

Enforce strict column data types to guarantee calculation accuracy.

BEST PRACTICES

Best Practices for Formulas & Power Query

Essential rules to follow for error-free, automated Excel reporting models.

✓ Always clean and prepare data in Power Query BEFORE applying complex formulas.

✓ Always specify exact match (FALSE or 0) when using VLOOKUP or HLOOKUP.

✓ Prefer XLOOKUP over VLOOKUP whenever Excel 365 / 2021 is available.

✓ Wrap risky lookup formulas with IFERROR or IFNA to prevent #N/A display.

✓ Document complex nested formulas with column header notes for team clarity.

— SUMMARY

Key Takeaways from Session 05

Core principles to retain as you progress toward advanced analytics mastery:



01

Saves Hours of Manual Work

Text functions & Power Query eliminate tedious line-by-line cell cleaning.



02

Automates Repetitive Workflows

Click Refresh to run recorded ETL transformations on new monthly data files.



03

Improves Data Quality

Enforces strict data types, removes whitespace, and strips duplicates automatically.



04

Flexible Lookup Architecture

Mastering XLOOKUP and INDEX/MATCH enables lookups in any direction across tables.



05

Dynamic Array Power

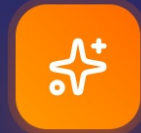
FILTER, UNIQUE, and SORT create live, auto-expanding reports without VBA.



06

Essential Data Analyst Skill

Power Query is the cornerstone of modern Excel, Power BI, and SQL analytics pipelines.



From Formula to Insight.

Every function and query transformation you learned today turns a spreadsheet column into a decision someone can act on.

Next session: Power Query Deep Dive (Merge, Append, Unpivot & Group By)

Instant Academy • Data Analysis Training Program